

9(a)	time for activity (of sample) to halve	B1
9(b)	sketch: line with positive gradient starting at (0,0) and extending to $t = 80$ min	B1
	exponential curve, extending from $t = 0$ to $t = 80$ min, with gradient of steadily decreasing magnitude	B1
	line passing through (0,0), (20, $0.5N_0$) and (40, $0.75 N_0$)	B1
9(c)(i)	every (undecayed) nucleus has the same probability of decay	M1
	fewer (undecayed) nuclei remaining (with time), so fewer will decay (in a given time interval)	A1
9(c)(ii)	- sample emits in all directions but detector only captures emissions in one direction - some emissions are absorbed before reaching detector - some emissions are scattered within the sample - simultaneous arrival of multiple particles only registers once - some particles may reach detector but not cause ionisation <i>Any two points, 1 mark each</i>	B2
	measured count rate is less than the activity	B1